

DEPARTMENT OF INFORMATION TECHNOLOGY

24ITT32 - PROGRAMMING IN JAVA

INDIVIDUAL PROJECT ON
ACADEMIC MANAGEMENT SYSTEM

P R O J E C T R E P O R T

Submitted by

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25ITR169

JAVA ASSIGNMENT

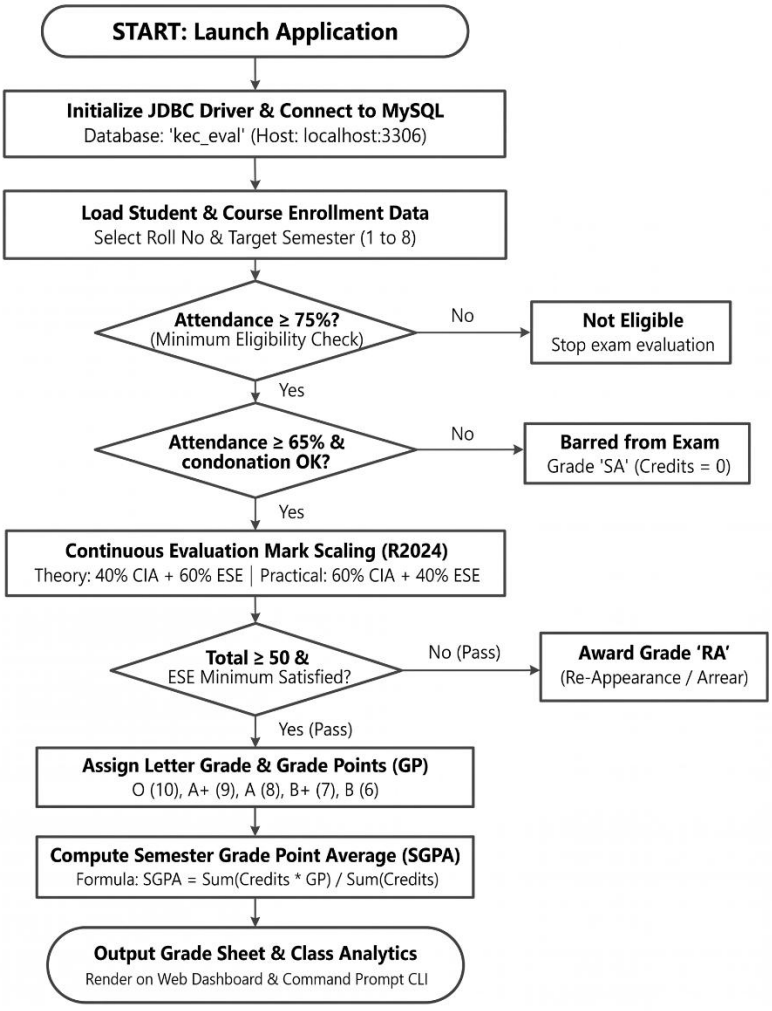
AIM:

Development of Kongu Engineering College (R2024) Student Academic & Continuous Evaluation System using Java JDBC.

OBJECTIVE:

Design and develop a database-driven Java application using JDBC to manage student records, perform incremental marks entry (Continuous Internal Assessments and End Semester Examinations), enforce attendance eligibility rules, and generate formal academic performance reports according to Kongu Engineering College (Autonomous) - Regulation 2024. System Requirements & Specifications.

FLOW CHART:



PROCEDURE:

- Load the MySQL Connector/J driver to enable communication between the Java application and the MySQL database.

- Establish a connection to the kec_eval database running on the local MySQL server using DriverManager.getConnection().

- Store and retrieve student details, including roll number, name, department, and semester, along with course information such as course code, title, credits, and course type.

- Enroll students in semester courses, record the number of classes held and attended, and calculate the attendance percentage.

- Verify examination eligibility based on attendance percentage. Students with 75% or above are eligible, those with 65%–74% require medical condonation verification, and those below 65% are marked as Shortage of Attendance (SA).

- Record CIA-1, CIA-2, and assignment marks, and calculate the scaled internal assessment marks according to the specified theory and practical course weightages.

- Record end-semester examination marks out of 100, verify the minimum passing requirements, and scale the marks according to the applicable theory and practical .

- Calculate the total marks by combining scaled CIA and ESE marks. Assign letter grades and grade points according to the specified grading scale, marking students as RA if they fail the examination or score below the required passing mark.

- Generate student grade sheets, attendance shortage reports, and department-wise pass percentage reports. Display the results through the interactive web dashboard and the command prompt console**.**

DATABASE SCHEMA:

1.STUDENTS TABLE:

```
mysql> describe students;
+------------------+-------------+------+-----+-------------------+-------------------+
| Field            | Type        | Null | Key | Default           | Extra             |
+------------------+-------------+------+-----+-------------------+-------------------+
| roll_no          | varchar(20) | NO   | PRI | NULL              |                   |
| name             | varchar(100)| NO   |     | NULL              |                   |
| department       | varchar(20) | NO   |     | NULL              |                   |
| current_semester | int         | NO   |     | 3                 |                   |
| created_at       | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+-------------+------+-----+-------------------+-------------------+
5 rows in set (0.03 sec)
```

2. COURSES TABLE:

```
mysql> describe courses;
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
| Field       | Type                              | Null | Key | Default           | Extra             |
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
| course_code | varchar(20)                       | NO   | PRI | NULL              |                   |
| title       | varchar(150)                      | NO   |     | NULL              |                   |
| credits     | int                               | NO   |     | NULL              |                   |
| course_type | enum('THEORY','PRACTICAL','EMBEDDED') | NO   |     | NULL              |                   |
| semester    | int                               | NO   |     | 3                 |                   |
| created_at  | timestamp                         | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+-------------+-----------------------------------+------+-----+-------------------+-------------------+
6 rows in set (0.01 sec)
```

3. ENROLLMENTS TABLE:

```
mysql> describe enrollments;
+------------------+-------------+------+-----+-------------------+-------------------+
| Field            | Type        | Null | Key | Default           | Extra             |
+------------------+-------------+------+-----+-------------------+-------------------+
| enrollment_id    | int         | NO   | PRI | NULL              | auto_increment    |
| roll_no          | varchar(20) | NO   | MUL | NULL              |                   |
| course_code      | varchar(20) | NO   | MUL | NULL              |                   |
| semester         | int         | NO   |     | 3                 |                   |
| total_classes    | int         | NO   |     | 0                 |                   |
| attended_classes | int         | NO   |     | 0                 |                   |
| is_condoned      | tinyint(1)  | NO   |     | 0                 |                   |
| created_at       | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
+------------------+-------------+------+-----+-------------------+-------------------+
8 rows in set (0.01 sec)
```

4. ASSESSMENT MARKS TABLE:

```
mysql> describe assessment_marks;
+-----------------+-------------+------+-----+-----------------+-----------------------------------------+
| Field           | Type        | Null | Key | Default         | Extra                                   |
+-----------------+-------------+------+-----+-----------------+-----------------------------------------+
| mark_id         | int         | NO   | PRI | NULL            | auto_increment                          |
| enrollment_id   | int         | NO   | UNI | NULL            |                                         |
| cia1_marks      | decimal(5,2) | YES  |     | NULL            |                                         |
| cia2_marks      | decimal(5,2) | YES  |     | NULL            |                                         |
| assignment_marks | decimal(5,2) | YES  |     | NULL            |                                         |
| ese_marks       | decimal(5,2) | YES  |     | NULL            |                                         |
| is_ese_absent   | tinyint(1)  | NO   |     | 0               |                                         |
| updated_at      | timestamp   | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED on update CURRENT_TIMESTAMP |
+-----------------+-------------+------+-----+-----------------+-----------------------------------------+
8 rows in set (0.02 sec)
```

SOURCE CODE:

MAINAPP.JAVA:

```
import java.sql.SQLException;
import java.util.List;
import java.util.Map;
import java.util.Scanner;

/**
 * MainApp - Menu-Driven Command Line Interface for Kongu Engineering College (R2024)
 * Student Academic & Continuous Evaluation System using Java JDBC.
 */
public class MainApp {

    private static final StudentDAO studentDAO = new StudentDAO();
    private static final AcademicDAO academicDAO = new AcademicDAO();
    private static final Scanner scanner = new Scanner(System.in);

    public static void main(String[] args) {
        // Support automated demo runner flag for testing and screen capture
        if (args.length > 0 && ("--demo".equalsIgnoreCase(args[0]) || "--
                showcase".equalsIgnoreCase(args[0]))) {
            runAutomatedDemonstration();
            return;
        }

        printBanner();
        boolean running = true;
        while (running) {
            printMainMenu();
            String choice = promptString("Enter your choice: ");
            System.out.println();
            switch (choice.trim()) {
                case "1":
                    handleStudentMenu();
                    break;
                case "2":
                    handleCourseMenu();
                    break;
                case "3":
                    handleEnrollmentMenu();
                    break;
                case "4":
                    handleIncrementalMarksMenu();
                    break;
                case "5":
                    handleReportsMenu();
                    break;
                case "6":
                    handleUtilitiesMenu();
```

```
                break;
            case "7":
                runAutomatedDemonstration();
                break;
            case "0":
                System.out.println("Exiting Kongu Engineering College Evaluation System.
                        Goodbye!");
                running = false;
                break;
            default:
                System.out.println("[!] Invalid choice. Please select from the menu
                        options.");
        }
        if (running) {
            System.out.println("\nPress ENTER to continue...");
            scanner.nextLine();
        }
    }
}

// ========================================================================
// UI HEADERS & MENUS
// ========================================================================

private static void printBanner() {
    System.out.println("========================================================================
            ========================");
    System.out.println("           KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION
            2024");
    System.out.println("             STUDENT ACADEMIC & CONTINUOUS EVALUATION SYSTEM
            (JDBC)");
    System.out.println("========================================================================
            ========================");
}

private static void printMainMenu() {
    System.out.println("\n================================== MAIN MENU
            ==================================");
    System.out.println(" [1] Student Management (Register, View, Search)");
    System.out.println(" [2] Course Catalog (Add Course, View Catalog)");
    System.out.println(" [3] Course Enrollment & Attendance Tracking (Enroll, Record
            Attendance)");
    System.out.println(" [4] Incremental Marks Entry (\"Then and Now\" - Phase 1, 2,
            3)");
    System.out.println(" [5] Reports & Academic Analytics (Mark Sheet, Defaulters,
            Summary)");
    System.out.println(" [6] Database Utilities (Test Connection, Re-seed Sample Data)");
    System.out.println(" [7] Run Automated Step-by-Step Demonstration (End-to-End
            Showcase)");
    System.out.println(" [0] Exit Application");
    System.out.println("========================================================================
            =============");
}

// ========================================================================
// 1. STUDENT MANAGEMENT
// ========================================================================

private static void handleStudentMenu() {
    System.out.println("--- STUDENT MANAGEMENT ---");
    System.out.println("1. Register New Student");
    System.out.println("2. View All Registered Students");
    System.out.println("3. Search Student by Roll Number");
    String ch = promptString("Select option (1-3): ");

    switch (ch) {
        case "1":
            registerStudentInteractive();
            break;
        case "2":
            viewAllStudents();
            break;
        case "3":
            searchStudent();
            break;
        default:
            System.out.println("[!] Invalid option.");
```

```
    }
}

private static void registerStudentInteractive() {
    System.out.println("\n--- [Student Registration] ---");
    String roll = promptString("Enter Roll Number (e.g. 24ITR010): ");
    String name = promptString("Enter Student Name: ");
    String dept = promptString("Enter Department (e.g. IT, CSE, ECE, MECH): ");
    int sem = promptInt("Enter Current Semester (1-8): ");

    try {
        Student existing = studentDAO.getStudent(roll);
        if (existing != null) {
            System.out.println("[!] Error: Student with Roll Number " + roll + " already
                    exists!");
            return;
        }
        Student s = new Student(roll, name, dept, sem);
        boolean ok = studentDAO.addStudent(s);
        if (ok) {
            System.out.println("[SUCCESS] Student successfully registered: " + s);
        } else {
            System.out.println("[!] Failed to register student.");
        }
    } catch (SQLException e) {
        System.err.println("[SQL ERROR] " + e.getMessage());
    }
}

private static void viewAllStudents() {
    try {
        List<Student> students = studentDAO.getAllStudents();
        System.out.println("\n--- [Registered Students Catalog] --- (" + students.size()
                + " Students)");
        System.out.printf("%-12s | %-25s | %-8s | %-8s\n", "Roll No", "Student Name",
                "Dept", "Semester");

                System.out.println("-----------------------------------------------------
                --------------");
        for (Student s : students) {
            System.out.printf("%-12s | %-25s | %-8s | %-8d\n",
                    s.getRollNumber(), s.getName(), s.getDepartment(),
                            s.getCurrentSemester());
        }
    } catch (SQLException e) {
        System.err.println("[SQL ERROR] " + e.getMessage());
    }
}

private static void searchStudent() {
    String roll = promptString("Enter Roll Number to Search: ");
    try {
        Student s = studentDAO.getStudent(roll);
        if (s != null) {
            System.out.println("\n[FOUND] Student Details:");
            System.out.println("  Roll Number     : " + s.getRollNumber());
            System.out.println("  Name            : " + s.getName());
            System.out.println("  Department      : " + s.getDepartment());
            System.out.println("  Current Semester: " + s.getCurrentSemester());
        } else {
            System.out.println("[!] No student found with Roll Number: " + roll);
        }
    } catch (SQLException e) {
        System.err.println("[SQL ERROR] " + e.getMessage());
    }
}
```

DB CONNECTION.JAVA:

```
import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;


/**

 * DBConnection - Database Connection Management for KEC Evaluation System

 * Kongu Engineering College (Autonomous) - Regulation 2024

 */

public class DBConnection {

    private static final String DEFAULT_HOST = "localhost";

    private static final String DEFAULT_PORT = "3306";

    private static final String DEFAULT_DB = "kec_eval";

    private static final String DEFAULT_USER = "root";

    private static final String DEFAULT_PASS = "0000";


    public static String getUrl() {

        String host = System.getenv("DB_HOST") != null ? System.getenv("DB_HOST") : DEFAULT_HOST;

        String port = System.getenv("DB_PORT") != null ? System.getenv("DB_PORT") : DEFAULT_PORT;

        String db = System.getenv("DB_NAME") != null ? System.getenv("DB_NAME") : DEFAULT_DB;

                    return "jdbc:mysql://" + host + ":" + port + "/" + db +
"?useSSL=false&allowPublicKeyRetrieval=true&serverTimezone=UTC";

    }


    public static String getUser() {

        return System.getenv("DB_USER") != null ? System.getenv("DB_USER") : DEFAULT_USER;

    }
```

```
    public static String getPassword() {
            return System.getenv("DB_PASSWORD") != null ? System.getenv("DB_PASSWORD") :
DEFAULT_PASS;
    }

    static {
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
        } catch (ClassNotFoundException e) {
            System.err.println("[ERROR] MySQL JDBC Driver not found in classpath: " + e.getMessage());
        }
    }

    /**
     * Obtains a new database connection
     */
    public static Connection getConnection() throws SQLException {
        return DriverManager.getConnection(getUrl(), getUser(), getPassword());
    }

    /**
     * Tests the database connectivity
     */
    public static boolean testConnection() {
        try (Connection conn = getConnection()) {
            return conn != null && !conn.isClosed();
        }
```

OUTPUT SCREENSHOTS:

MAIN MENU:

```
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\VETRI VEL MURUGAN>cd /d "c:\Users\VETRI VEL MURUGAN\Desktop\KECEval"

c:\Users\VETRI VEL MURUGAN\Desktop\KECEval>run.bat
Starting Kongu Engineering College Evaluation System...
================================================================================
           KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024
             STUDENT ACADEMIC & CONTINUOUS EVALUATION SYSTEM (JDBC)
================================================================================

================================== MAIN MENU ==================================
 [1] Student Management (Register, View, Search)
 [2] Course Catalog (Add Course, View Catalog)
 [3] Course Enrollment & Attendance Tracking (Enroll, Record Attendance)
 [4] Incremental Marks Entry ("Then and Now" - Phase 1, 2, 3)
 [5] Reports & Academic Analytics (Mark Sheet, Defaulters, Summary)
 [6] Database Utilities (Test Connection, Re-seed Sample Data)
 [7] Run Automated Step-by-Step Demonstration (End-to-End Showcase)
 [0] Exit Application
```

STUDENT & COURSE MANAGEMENT:

1.STUDENT REGISTRATION:

```
--- STUDENT MANAGEMENT ---
1. Register New Student
2. View All Registered Students
3. Search Student by Roll Number
Select option (1-3): 1

--- [Student Registration] ---
Enter Roll Number (e.g. 24ITR010): 25ITR158
Enter Student Name: SURIYA
Enter Department (e.g. IT, CSE, ECE, MECH): IT
Enter Current Semester (1-8): 3
[SUCCESS] Student successfully registered: [25ITR158] SURIYA              | Dept: IT    | Sem: 3
```

2. COURSE CATALOG:

```
--- COURSE CATALOG ---
1. Add New Course
2. View All Courses
Select option (1-2): 1

--- [Add New Course] ---
Enter Course Code (e.g. 24ITT35): 24ITC32
Enter Course Title: FULL STACK DEVELOPMENT
Enter Credits (e.g. 3.0, 4.0, 1.5): 3.0
Select Course Type:
  1. THEORY    (CIA: 40% | ESE: 60% | Min ESE: 45%)
  2. PRACTICAL (CIA: 60% | ESE: 40% | Min ESE: 50%)
  3. EMBEDDED  (CIA: 50% | ESE: 50% | Min ESE: 45%)
Choice (1-3): 3
Enter Semester (1-8): 2
[SUCCESS] Course successfully added: [24ITC32] FULL STACK DEVELOPMENT            | Credits: 3.0 | Type: EMBEDDED  | Sem: 2

Press ENTER to continue...
```

3. COURSE ENROLLMENT & ATTENDANCE TRACKING:

```
********************************************************************************
       KONGU ENGINEERING COLLEGE (AUTONOMOUS) - REGULATION 2024 DEMONSTRATION
                FULL END-TO-END AUTOMATED VERIFICATION SHOWCASE
********************************************************************************

>>> STEP 1: Adding a New Student and Course Catalog Entry...
[OK] Demo Student Exists: [24ITR099] Vigneshwaran S       | Dept: IT    | Sem: 3
[OK] Demo Course Exists: [24ITT39] Cloud Computing & DevOps           | Credits: 3.0 | Type: THEORY    | Sem: 3

>>> STEP 2: Enrolling Student into Course & Tracking Attendance (54/60 = 90.0%)...
[OK] Using existing enrollment #29 (Attendance: 90.0% - ELIGIBLE)
```

INCREMENTAL MARKS ENTRY:

1.PHASE 1 (MID-SEMESTER):

```
>>> STEP 3: Phased / Incremental Marks Entry ("Then and Now"):
  --> [Phase 1 (Mid-Semester)]: Entering CIA Test 1 = 88.0, CIA Test 2 = 92.0
      [Phase 1 SAVED successfully]
```

2.PHASE 2 (END OF SEMESTER - CIA):

```
  --> [Phase 2 (End of Semester - CIA)]: Entering Assignments / Quizzes = 94.0
      [Phase 2 SAVED successfully]
```

3. PHASE 3 (POST SEMESTER):

```
  --> [Phase 3 (Post Semester - ESE)]: Entering End Semester Exam Marks = 90.0
      [Phase 3 SAVED successfully]
```

ATTENDANCE & GRADING RULES:

1.ATTENDANCE FILTER:

```
>>> STEP 4: Demonstrating Attendance Filter & Condonation Rule Enforcement:
  Attempting to enter Phase 3 ESE marks for Attendance Defaulter (24ITR004 i
n 24ITT31, 55% attendance)...
  [RULE ENFORCED!] Engine blocked ESE entry: ATTENDANCE VIOLATION: Student i
s BARRED from ESE. Attendance shortage (55.0% < 65% threshold). Grade SA is
awarded.
```

```
>>> STEP 7: Attendance Defaulter Report (All students <65% flagged SA):
=================================================================================================
                        KONGU ENGINEERING COLLEGE (AUTONOMOUS)
                    ATTENDANCE DEFAULTER REPORT (BARRED - GRADE: SA)
                              SEMESTER: 3 | DEPARTMENT: ALL DEPARTMENTS
=================================================================================================
Roll No    | Student Name          | Dept   | Course    | Held / Attd  | Attd %   | Condoned   | Awarded Grade
-------------------------------------------------------------------------------------------------
24CSR003   | Jeevitha K            | CS     | 24CST31   |   50 / 28    |   56.0% | NO         | SA (BARRED)
24ITR004   | Dinesh Kumar P        | IT     | 24ITT31   |   60 / 33    |   55.0% | NO         | SA (BARRED)
24ITR004   | Dinesh Kumar P        | IT     | 24ITT32   |   60 / 32    |   53.3% | NO         | SA (BARRED)
24ITR005   | Elango R              | IT     | 24ITT31   |   60 / 41    |   68.3% | NO         | SA (BARRED)
-------------------------------------------------------------------------------------------------
  Total Defaulter Cases Flagged: 4  |  Policy: Shortage <65% or Unapproved Condonation = SA Grade
=================================================================================================
```

STUDENT SEMESTER MARK SHEET:

```
>>> STEP 5: Formal Student Semester Mark Sheet with SGPA & Percentage (24ITR001):
=================================================================================================================
                            KONGU ENGINEERING COLLEGE (AUTONOMOUS), PERUNDURAI
                               STUDENT SEMESTER GRADE SHEET - REGULATION 2024
=================================================================================================================
  Roll Number : 24ITR001              Student Name : Aadhavan K
  Department  : IT                    Semester     : 3
-----------------------------------------------------------------------------------------------------------------
Code      | Course Title                    | Type       | Cred | Attd%  | CIA1  | CIA2  | Asgn  | CIA-Sc | ESE   | ESE-Sc | Total | Grd  | GP | Status
-----------------------------------------------------------------------------------------------------------------
24ITI31   | Digital Principles and System D | EMBEDDED   |  4.0 |  91.7% |    90 |    86 |    95 |   45.4 |    88 |   44.0 |  89.4 | A+   |  9 | PASS
24ITL31   | Data Structures & Java Laborato | PRACTICAL  |  1.5 | 100.0% |    96 |    94 |    98 |   57.7 |    95 |   38.0 |  95.7 | O    | 10 | PASS
24ITT31   | Data Structures and Algorithms  | THEORY     |  3.0 |  93.3% |    95 |    92 |    98 |   38.1 |    94 |   56.4 |  94.5 | O    | 10 | PASS
24ITT32   | Object Oriented Programming wit | THEORY     |  3.0 |  96.7% |    88 |    90 |    92 |   36.1 |    91 |   54.6 |  90.7 | O    | 10 | PASS
-----------------------------------------------------------------------------------------------------------------
  Total Enrolled Credits : 11.5               Total Credits Earned    : 11.5
  Cumulative Grade Points: 111.00             Semester SGPA           : 9.65 / 10.00
  Equivalent Percentage  : 96.52 %            Overall Academic Result : PASS
=================================================================================================================


>>> STEP 6: Generated Mark Sheet for Newly Added Student (24ITR099):
=================================================================================================================
                           KONGU ENGINEERING COLLEGE (AUTONOMOUS), PERUNDURAI
                               STUDENT SEMESTER GRADE SHEET - REGULATION 2024
=================================================================================================================
  Roll Number : 24ITR099              Student Name : Vigneshwaran S
  Department  : IT                    Semester     : 3
-----------------------------------------------------------------------------------------------------------------
Code      | Course Title                    | Type       | Cred | Attd%  | CIA1  | CIA2  | Asgn  | CIA-Sc | ESE   | ESE-Sc | Total | Grd  | GP | Status
-----------------------------------------------------------------------------------------------------------------
24ITI31   | Digital Principles and System D | EMBEDDED   |  4.0 |  93.3% |    85 |    88 |    90 |   44.0 |    86 |   43.0 |  87.0 | A+   |  9 | PASS
24ITL31   | Data Structures & Java Laborato | PRACTICAL  |  1.5 |  96.7% |    92 |    95 |    96 |   56.7 |    92 |   36.8 |  93.5 | O    | 10 | PASS
24ITT39   | Cloud Computing & DevOps        | THEORY     |  3.0 |  90.0% |    88 |    92 |    94 |   36.6 |    90 |   54.0 |  90.6 | O    | 10 | PASS
-----------------------------------------------------------------------------------------------------------------
  Total Enrolled Credits : 8.5                Total Credits Earned    : 8.5
  Cumulative Grade Points: 81.00              Semester SGPA           : 9.53 / 10.00
  Equivalent Percentage  : 95.29 %            Overall Academic Result : PASS
=================================================================================================================
```

DEPARTMENT WISE CLASS PERFORMANCE & RESULT SUMMARY:

```
>>> STEP 8: Class Performance & Result Summary for Department of IT:
=============================================================================================
                        KONGU ENGINEERING COLLEGE (AUTONOMOUS)
                    CLASS PERFORMANCE & RESULT SUMMARY - SEMESTER 3
                                DEPARTMENT: IT
=============================================================================================
  Total Students Evaluated  : 9
  Students Passed (Clear All): 4
  Students with Arrears     : 4     (Attendance Barred SA: 2)
  Department Pass Percentage: 44.44%
  Class Average SGPA        : 5.84 / 10.00 (Equivalent: 58.38%)
---------------------------------------------------------------------------------------------
Grade Distribution Across All Course Assessments:
  O  : 5   | A+ : 3   | A  : 3   | B+ : 6   | B  : 3   | RA : 1   | SA : 3   | AB : 1
---------------------------------------------------------------------------------------------
Academic Merit / Rank List:
Rank | Roll Number  | Student Name             | SGPA   | Equiv %  | Result Standing
---------------------------------------------------------------------------------------------
1    | 24ITR001     | Aadhavan K               | 9.65   | 96.52  % | PASS
2    | 24ITR099     | Vigneshwaran S           | 9.53   | 95.29  % | PASS
3    | 24ITR002     | Bavithra S               | 7.52   | 75.22  % | PASS
4    | 24ITR003     | Chandru M                | 6.52   | 65.22  % | PASS
5    | 24ITR006     | Farhana A                | 3.50   | 35.00  % | RE-APPEARANCE (ARREAR)
6    | 24ITR007     | Gokulnath V              | 3.50   | 35.00  % | RE-APPEARANCE (ARREAR)
7    | 24ITR004     | Dinesh Kumar P           | 3.48   | 34.78  % | SHORTAGE OF ATTENDANCE (SA)
8    | 24ITR005     | Elango R                 | 3.00   | 30.00  % | SHORTAGE OF ATTENDANCE (SA)
=============================================================================================
```

DATABASE:

STUDENTS DATABASE:

```
mysql> USE kec_eval;
Database changed
mysql> SELECT * FROM students;
+----------+--------------------+------------+------------------+---------------------+
| roll_no  | name               | department | current_semester | created_at          |
+----------+--------------------+------------+------------------+---------------------+
| 22ADR071 | Sudharsan G        | AIDS       |                7 | 2026-10-19 21:29:39 |
| 22ADR081 | Akilesh R          | AIDS       |                8 | 2026-10-19 21:29:39 |
| 22CER071 | Saranya V          | CIVIL      |                7 | 2026-10-19 21:29:39 |
| 22CER081 | Yogeshwaran T      | CIVIL      |                8 | 2026-10-19 21:29:39 |
| 22CHR071 | Swetha T           | CHEM       |                7 | 2026-10-19 21:29:39 |
| 22CHR081 | Balamurugan S      | CHEM       |                8 | 2026-10-19 21:29:39 |
| 22CSR071 | Praveen Kumar R    | CSE        |                7 | 2026-10-19 21:29:39 |
| 22CSR072 | Priyanga M         | CSE        |                7 | 2026-10-19 21:29:39 |
| 22CSR081 | Venkatesh S        | CSE        |                8 | 2026-10-19 21:29:39 |
| 22CSR082 | Vidhya R           | CSE        |                8 | 2026-10-19 21:29:39 |
| 22ECR071 | Rajesh K           | ECE        |                7 | 2026-10-19 21:29:39 |
| 22ECR081 | Vigneshwar B       | ECE        |                8 | 2026-10-19 21:29:39 |
| 22EER071 | Rithika S          | EEE        |                7 | 2026-10-19 21:29:39 |
| 22EER081 | Vishnu Vardhan M   | EEE        |                8 | 2026-10-19 21:29:39 |
| 22ITR071 | Manikandan P       | IT         |                7 | 2026-10-19 21:29:39 |
| 22ITR072 | Nandhini S         | IT         |                7 | 2026-10-19 21:29:39 |
```

COURSES DATABASE:

```
mysql> SELECT * FROM courses;
+-------------+-----------------------------------------------+---------+-------------+----------+---------------------+
| course_code | title                                         | credits | course_type | semester | created_at          |
+-------------+-----------------------------------------------+---------+-------------+----------+---------------------+
| 24ADC31     | Artificial Intelligence & Knowledge Rep.      |       4 | EMBEDDED    |        3 | 2026-10-19 20:59:10 |
| 24ADC41     | Machine Learning Algorithms & Systems         |       4 | EMBEDDED    |        4 | 2026-10-19 20:59:10 |
| 24ADC51     | Natural Language Processing (NLP)             |       4 | EMBEDDED    |        5 | 2026-10-19 20:59:10 |
| 24ADC61     | Generative AI & Large Language Models         |       4 | EMBEDDED    |        6 | 2026-10-19 20:59:10 |
| 24ADC71     | Reinforcement Learning & Autonomous Agents    |       4 | EMBEDDED    |        7 | 2026-10-19 20:59:10 |
| 24ADL31     | Python for AI & Data Science Laboratory       |       1 | PRACTICAL   |        3 | 2026-10-19 20:59:10 |
| 24ADL41     | Machine Learning & Data Mining Lab            |       1 | PRACTICAL   |        4 | 2026-10-19 20:59:10 |
| 24ADL51     | Deep Learning & NLP Laboratory                |       1 | PRACTICAL   |        5 | 2026-10-19 20:59:10 |
| 24ADL61     | Computer Vision & GenAI Laboratory            |       1 | PRACTICAL   |        6 | 2026-10-19 20:59:10 |
| 24ADP71     | AI Capstone Project Phase I                   |       3 | PRACTICAL   |        7 | 2026-10-19 20:59:10 |
| 24ADP81     | AI Capstone Project Phase II / Industry Track |      10 | PRACTICAL   |        8 | 2026-10-19 20:59:10 |
| 24ADT31     | Mathematical Foundations for Data Science     |       4 | THEORY      |        3 | 2026-10-19 20:59:10 |
| 24ADT41     | Statistical Learning & Data Mining            |       4 | THEORY      |        4 | 2026-10-19 20:59:10 |
| 24ADT51     | Deep Learning & Neural Networks               |       4 | THEORY      |        5 | 2026-10-19 20:59:10 |
| 24ADT61     | Computer Vision & Video Analytics             |       4 | THEORY      |        6 | 2026-10-19 20:59:10 |
| 24ADT71     | AI Ethics, Governance & MLOps Infrastructure  |       3 | THEORY      |        7 | 2026-10-19 20:59:10 |
| 24CEC31     | Surveying and Geomatics                       |       4 | EMBEDDED    |        3 | 2026-10-19 20:59:10 |
| 24CEC41     | Fluid Mechanics & Hydraulic Machinery         |       4 | EMBEDDED    |        4 | 2026-10-19 20:59:10 |
| 24CEC51     | Soil Mechanics & Geotechnical Engineering     |       4 | EMBEDDED    |        5 | 2026-10-19 20:59:10 |
| 24CEC61     | Environmental Engineering & Water Treatment   |       4 | EMBEDDED    |        6 | 2026-10-19 20:59:10 |
| 24CEC71     | Transportation Engineering & GIS              |       4 | EMBEDDED    |        7 | 2026-10-19 20:59:10 |
| 24CEL31     | Surveying Field Work Laboratory               |       1 | PRACTICAL   |        3 | 2026-10-19 20:59:10 |
| 24CEL41     | Materials Testing & Hydraulics Lab            |       1 | PRACTICAL   |        4 | 2026-10-19 20:59:10 |
| 24CEL51     | Geotechnical Engineering Lab                  |       1 | PRACTICAL   |        5 | 2026-10-19 20:59:10 |
| 24CEL61     | Environmental Engineering Laboratory          |       1 | PRACTICAL   |        6 | 2026-10-19 20:59:10 |
| 24CEP71     | Project Work Phase I                          |       3 | PRACTICAL   |        7 | 2026-10-19 20:59:10 |
```

ENROLLMENTS DATABASE:

```
mysql> SELECT * FROM enrollments;
+---------------+----------+-------------+----------+---------------+------------------+-------------+---------------------+
| enrollment_id | roll_no  | course_code | semester | total_classes | attended_classes | is_condoned | created_at          |
+---------------+----------+-------------+----------+---------------+------------------+-------------+---------------------+
|             1 | 24ITR001 | 24ITT31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|             2 | 24ITR001 | 24ITC31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|             3 | 24ITR001 | 24ITT32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|             4 | 24ITR001 | 24ITC32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|             5 | 24ITR001 | 24ITT33     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|             6 | 24ITR001 | 24MNT31     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|             7 | 24ITR001 | 24ITL31     |        3 |            30 |               26 |           0 | 2026-10-19 20:59:10 |
|             8 | 24ITR001 | 24ITL32     |        3 |            30 |               27 |           0 | 2026-10-19 20:59:10 |
|             9 | 24ITR001 | 24ITL33     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            10 | 24ITR002 | 24ITT31     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|            11 | 24ITR002 | 24ITC31     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
|            12 | 24ITR002 | 24ITT32     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            13 | 24ITR002 | 24ITC32     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            14 | 24ITR002 | 24ITT33     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            15 | 24ITR002 | 24MNT31     |        3 |            60 |               56 |           0 | 2026-10-19 20:59:10 |
|            16 | 24ITR002 | 24ITL31     |        3 |            30 |               27 |           0 | 2026-10-19 20:59:10 |
|            17 | 24ITR002 | 24ITL32     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            18 | 24ITR002 | 24ITL33     |        3 |            30 |               25 |           0 | 2026-10-19 20:59:10 |
|            19 | 24ITR003 | 24ITT31     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            20 | 24ITR003 | 24ITC31     |        3 |            60 |               53 |           0 | 2026-10-19 20:59:10 |
|            21 | 24ITR003 | 24ITT32     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            22 | 24ITR003 | 24ITC32     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            23 | 24ITR003 | 24ITT33     |        3 |            60 |               56 |           0 | 2026-10-19 20:59:10 |
|            24 | 24ITR003 | 24MNT31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|            25 | 24ITR003 | 24ITL31     |        3 |            30 |               28 |           0 | 2026-10-19 20:59:10 |
|            26 | 24ITR003 | 24ITL32     |        3 |            30 |               25 |           0 | 2026-10-19 20:59:10 |
|            27 | 24ITR003 | 24ITL33     |        3 |            30 |               26 |           0 | 2026-10-19 20:59:10 |
|            28 | 24CSR001 | 24CST31     |        3 |            60 |               49 |           0 | 2026-10-19 20:59:10 |
|            29 | 24CSR001 | 24CSC31     |        3 |            60 |               55 |           0 | 2026-10-19 20:59:10 |
|            30 | 24CSR001 | 24CST32     |        3 |            60 |               51 |           0 | 2026-10-19 20:59:10 |
```

RESULT:

Thus, the Student Academic and Continuous Evaluation System for Kongu Engineering College (Autonomous) Regulation 2024 was successfully designed, developed, and tested using Java (JDBC) and MySQL. The system accurately automates the calculation of Continuous Internal Assessments (CIA), End Semester Examinations (ESE), Letter Grades, Attendance Eligibility, and Semester Grade Point Average (SGPA).

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GITHUB:

LINK:	https://github.com/vetrivelmurugan-svg/JAVA-PROJECT.git